GSB

Exact String Matching

-> finding certain patterns that serve as recognition elements in biological processes, eg: transcription factor binding

Problem:

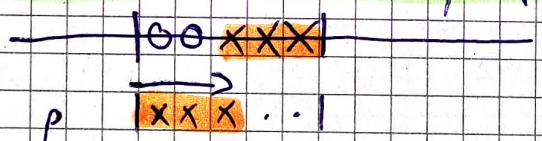
Given a string t, the problem of string matching deals with finding ~~where~~ whether a pattern p occurs in t and if p does occur then return position in t where p occurs.

-> Brute Force : $O(mn)$

Worst Case: $t = XXXX \dots X$, $p = XXXY$

-> Prefix Searching:

At position in the window we seek the longest suffix of the search window that is also a prefix of that pattern



-> Suffix Searching:

The search is conducted backwards along the search window, matching a suffix in the search window to a suffix of the pattern. By avoiding to read some characters this can lead to sub-linear average case algorithms.

-> Factor Searching:

The search is done ←, looking for the longest suffix in the window that is also a factor of the pattern

GSB

Knuth - Morris - Pratt (KMP)

-> prefix - based , left - to - right!

-> $O(m+n)$ Time

-> Idea:

If pattern fails to match, slide pattern to right by as many positions as possible without permitting a match to go unnoticed.

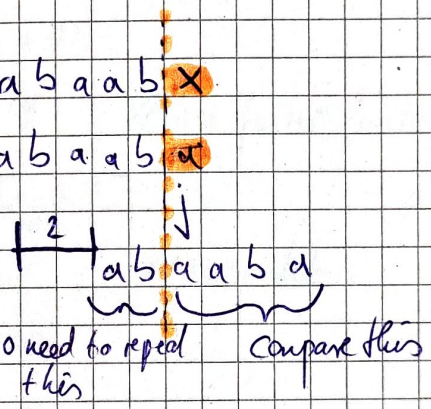
-> Position to shift:

The length of the largest prefix $P(0..j)$ that is a suffix of $P(1..j)$

eg:

t: ... a b a a b x

p: a b a a b x



-> Preprocess pattern p:

Failure function $F(j)$: the size of the largest prefix of $P(0..j)$ that is also a suffix of $P(1..j)$

-> Mismatch at position j:

$P[j] \neq T[j] \Rightarrow$ set $j \leftarrow F(j-1)$ to skip positions

GSD

Compute Failure Function (KMP)

→ $O(m)$ Time

```
F[0] ← 0
i ← 1
j ← 0
m ← length(P)
while i < m :
    if P[i] == P[j] :          // we have matched j+1 chars
        F[i] ← j+1
        i ← i+1
        j ← j+1
    else if j > 0 :            // use failure function to shift P
        j ← F[j-1]
    else :                     // no match
        F[i] ← 0
        i ← i+1
```

KMP Alg

→ O(m+n) Time

```
F ← failureFunction(P)
i ← 0
j ← 0
while i < n
    if T[i] == P[j]:
        if j == m-1:        // Match found
            return i-j
        else
            i ← i+1
            j ← j+1
    else
        if j > 0:
            j ← F[j-1]
        else
            i ← i+1
return -1
```

GrSB - Ex 4

1) Knuth - Morris - Pratt

pattern: ATGATTATGAATGA

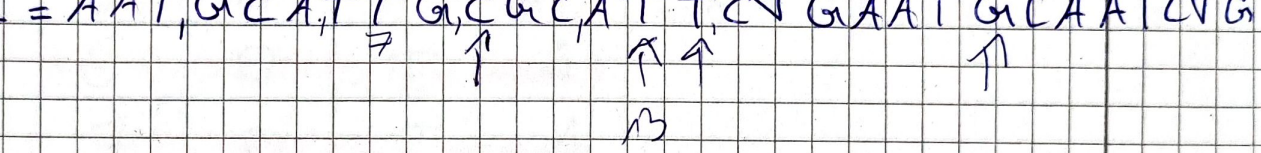
a)

j	0	1	2	3	4	5	6	7	8	9	10	11	12	13
P[j]	A	T	G	A	T	T	A	T	G	A	A	T	G	A
F[j]	0	0	0	1	2	0	1	2	3	4	1	2	3	4

b)

j	0	1	2	3	4	5	6	7
P[j]	A	A	T	G	C	A	A	T
F[j]	0	1	0	0	0	1	2	3

T = AAT,GCA,TTG,CGC,ATT,CTGAATGCAATCTG



GSB - Ex 4

a) KMP

b)

	i	j	T[i]	P[j]	
	0	0	A	A	
	1	1	A	A	
	2	2	T	T	
	3	3	G	G	
	4	4	C	C	
	5	5	A	A	
j>0	6	6	T	A	F(6-1) = 1
j>0	6	1	T	A	F(1-1) = 0
	6	0	T	A	
	7	0	T	A	
	8	0	G	A	
	9	0	C	A	
	10	0	G	A	
	11	0	C	A	
	12	0	A	A	
j>0	13	1	T	A	F(1-1) = 0
	13	0	T	A	
	14	0	T	A	
	15	0	C	A	
	16	0	T	A	
	17	0	G	A	
	18	0	A	A	
	19	1	A	A	
	20	2	T	T	

i	j	T[i]	P[j]
21	3	G	G
22	4	C	C
23	5	A	A
24	6	A	A
25	7	T	T

c) Match position: 18

d) # Iterations: 25

e) Worst Case

Text: abcea...abce (m-1 times a)

Pattern: aaaa...b (m times a)

Match all but not the last char.

Boyer - Moore - Alg.

→ $O(mn)$ Time but in practice fast!

→ right - to - left (suffix - based approach)

→ Much faster on long targets

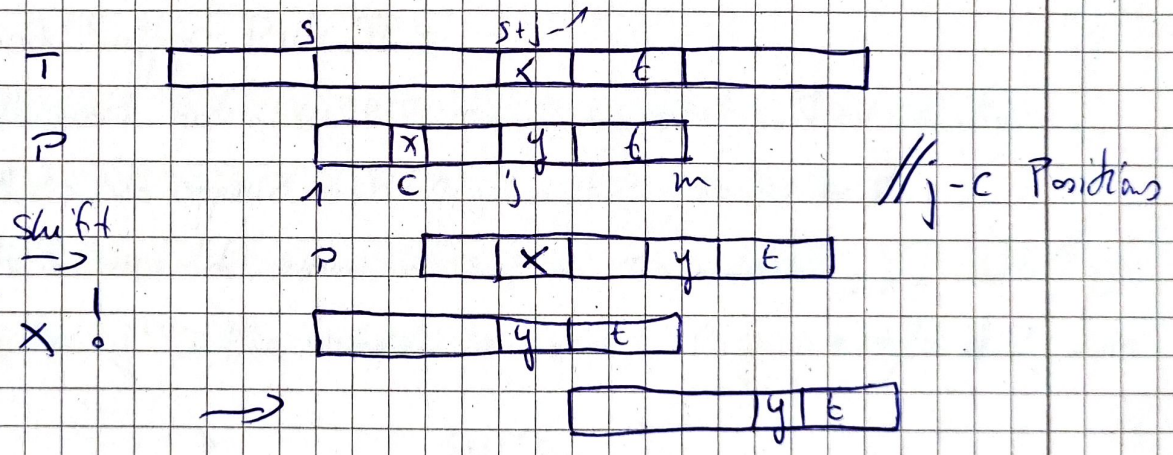
→ d smallest integer s.t.: $T[k+m-1] = p[m-1-d]$
$T[k+m-2] = p[m-2-d]$
$T[k+i] = p[i-d]$

→ k = starting index in target string

→ m : length of pattern

→ i : index of mismatch in pattern string

- Bad Character Rule : $\max(1, \text{len}(P) - \text{index_rightmost} - 1)$



→

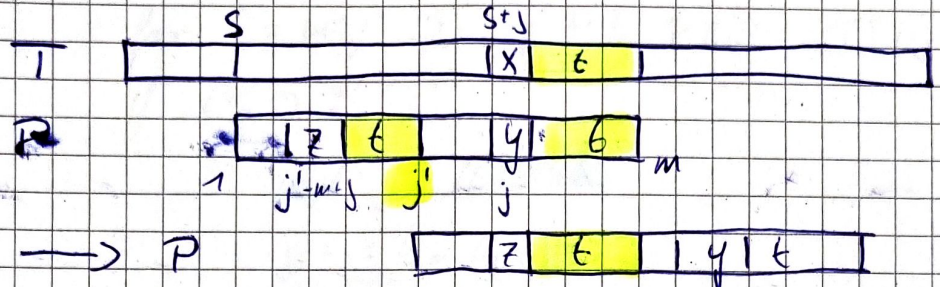
$b[T[i+j] - m + 1 + i]$

GSS

Good-Suffix-Rule (Boyer-Moore-Alg)

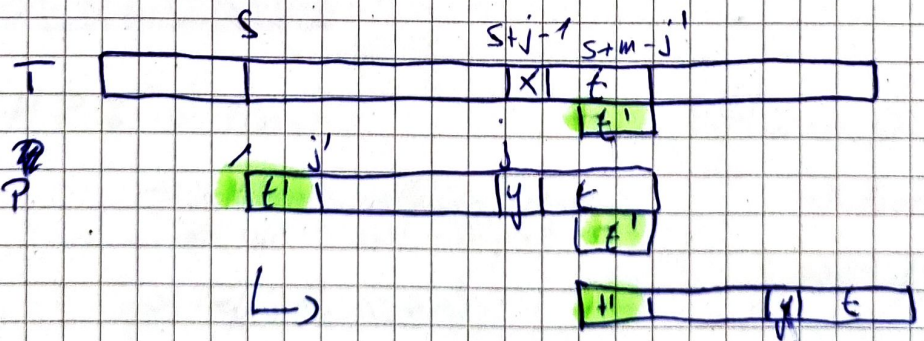
- **Good Suffix Rule I:**

  If a mismatch occurs in T_{s+j-1}, we match T_{s+j-1} with $P_{j'-m+j}$, where j' $(m-j+1 \leq j' < m)$ is the largest position s.t.:
  → $P_{j'+1,m}$ is a suffix of $P_{1,j'}$
  → $P_{j'-(m-j)} \neq P_j$



- **Good Suffix Rule II:**

  Is used when GSR-I can not be used. That is t does not appear in $P(1,j)$. t is unique in P.
  If a mismatch occurs in T_{s+j-1}, we match $T_{s+m-j'}$ with P_1, where j' is the largest position s.t. $P_{1,j'}$ is a suffix of $P_{j+1,m}$



GrSB

Boyer Moore Alg

```
bmGs ← good_suffix-rule (P)
bmBc ← bad_char-rule (P, alpha)        // preprocess

j ← 0                                  // Search
while j <= n - m :
    for i = m-1; i >= 0 && P[i] == T[i+j]; i--;
    if i < 0 :
        print j                        // Match
        j += bmGs[0]
    else :
        j += max(bmGs[i], bmBc[T[i+j]] - m + 1 + i)
```

- Preprocess: $O(m + |A|)$
- Worst Case: $O(mn)$, $T = A \ldots A$, $P = XAAA$
- $O(m+n)$ when p is not in T

KMP vs BM

KMP:
- → linear
- → Deals with repetitive strings as with other strings
- → Prefixed-Based
- → Even slower than Naive Alg in practice

BM:
- → On average sublinear
- → Problems with repetitions
- → Suffix-Based
- → Fastest in practice

⇒ Exakt String Matching is ~~easy~~ rare and often ask for in Biology

GSB - Ex 4

a) Boyer-Moore String Search

14 - 11 + 1

0 1 2 3 4 5 6 7 8 9 10 11 12 13

A T G A T T A T G . A A T G A

a)

Σ	A	T	G
B[j]	3	2	1

len(P) - max_id - 1

↳ 14 - 11 - 1 = 2

↳ 14 - 10 - 1 = 3

b)

j	0	1	2	3	4	5	6	7	8	9	10	11	12	13
P[j]	A	T	G	A	T	T	A	T	G	A	A	T	G	A
GS[j]	13	13	13	10	10	10	10	10	10	4	13	13	3	1

c)

→ Diff Bad Char Table

S: G C A T C G C A G A G A G T A T A C A G T A C G

P: G C A G A G A G // BC(A) - 8 + 1 + i = 1 - 8 + 1 + 8 - 1 = 1; GS = 1; j++

j=0

G C A G A G A G // BC(C) - 8 + 1 + i = 6 - 8 + 1 + 5 = 4; GS = 4; j += 4

j=1

. . . G C A G A G A G

→ 2 Shifts!

→ Second shift 4 chars!

d) BMH >> BM:

→ In BMH we use the last char of the suffix to perform the shift

→ In BMH we use do not use the GS rule but only the BC-rule

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Sequence Alignment

-> searching for seq features like binding sites
-> Mapping next-gen seq reads to reference genome
-> Building overlap graph for de novo assembly
-> Searching seq databases

- Global : alignment of two strings
- Local : Finding local similarities

-> pairwise -> lining up two seq's

- semi-global : approximative pattern matching
- free-end-gaps : Finding overlaps

Homology

-> Two proteins are homologous, when they share a common ancestor
-> Orthologs:

Homologous seq's in different species that arose from a common ancestral gene during speciation; may or may not be responsible for a similar function.

-> Paralogs:

Homologous seq's within a single species that arose by gene duplication

-> Identity : The extent to which two seq's are invariant
-> Conservation : Changes at a specific position of an AS or seq that preserve the physico-chemical properties of the original residue.
-> Similarity : The extent to which nucleotide or protein seq's are related. It is based upon identity plus conservation

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Pairwise Alignment

The process of lining up two seq's to achieve maximal level of identity for the purpose of assessing the degree of similarity and the possibility of homology.

→ Tool: BLAST

→ Protein Sequences can be more informative than DNA
↳ many amino acids share related biophysical properties.

↳ Codons:
changes in the third position of the triplet often do not alter the corresponding amino acid

→ DNA sequences can be more informative than protein
↳ If studying non-coding DNA

- Sequence Alignment:
  In: pair of seq's, score cost function
  Out: determine the correspondig between substrings in the seq's s.t. the similarity score is max.
  ↳ local similarities

Issues: • different length, • only small regions that match
• allow partial match,

- Sequence Variations:
  → Substitution: A C G A → A G G A
  gaps → Deletion: A C A → A C
  gaps → Insertion: A C → A G C

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Needleman Wunsch Alg

-> Dynamic Programming:
- Break problem into smaller subproblems
- Solve smaller problems optimal
- Use subproblems solution for optimal solution

-> can only be used when the property of overlapping subproblems. (Bellmans principle)

- **Step I: Initialization:**

1st row and 1st column of the score and traceback matrices,

Score & Tr are initialized:

Score		S	E	N	D
	0	g	2g	3g	4g
A	g				
N	2g				
D	3g				

Tr		S	E	N	D
		done	l	l	l
	A	u			
	N	u			
	D	u			

- **Step II: Calculation of scores:**

$X = x_1 \dots x_m$, $Y = y_1 \dots y_n$

$$S(i,j) = \max \begin{cases} S(i-1, j-1) + \tau(x_i, y_j) & , \ x_i \to y_j \\ S(i-1, j) + g & , \ x_i \to \varepsilon \\ S(i, j-1) + g & , \ \varepsilon \to y_j \end{cases}$$

τ: substitution score

g: gap penalty

- **Step III: Traceback:**

From (m, n) traceback (max) to (0, 0)

GSB - Ex 5

1) Needleman-Wunsch Alg

$g = -2$, match: $+1$, mismatch -1

Score		Y	C	T	Y	S	R	F	T	V	F
	0	-2	-4	-6	-8	-10	-12	-14	-16	-18	-20
Y	-2	1	-1	-3	-5	-7	-9	-11	-13	-15	-17
R	-4	-1	0	-2	-4	-6	-6	-8	-10	-12	-14
T	-6	-3	-2	1	-1	-3	-5	-7	-7	-9	-11
Y	-8	-5	-4	-1	2	0	-2	-4	-6	-8	-10
T	-10	-7	-6	-3	0	1	-1	-3	-3	-5	-7
V	-12	-9	-8	-5	-2	-1	0	-2	-4	-2	-4
F	-14	-11	-10	-7	-4	-3	-2	-1	-1	-3	-1

TB		Y	C	T	Y	S	R	F	T	V	F
	Point	l	l	l	l	l	l	l	l	l	l
Y	u	D	l	l	l	l	l	l	l	l	l
R	u	u	D	l	l	l	D	l	l	l	l
T	u	u	u	D	l	l	l	l	D	l	l
Y	u	u	u	u	D	l	l	l	l	l	l
T	u	u	u	u	u	D	l	l	D	l	l
V	u	u	u	u	u	u	D	l	l	D	l
F	u	u	u	u	u	u	u	D	l	l	D

$$\begin{pmatrix} Y & C & T & Y & S & R & F & T & V & F \\ Y & R & T & Y & - & - & - & T & V & F \end{pmatrix}$$

Local Alignment

-> capture seq's that share a common motif or domain but differ elsewhere
-> More sensitive when comparing highly diverged seq's

Smith-Waterman Alg

-> Comparison of alignments of any possible length starting and ending at any position in two seq's.
-> Guarantee to find optimal local alignment
-> substitution matrix & gap-scoring scheme

I) Initialization: 1st rows & 1st colum. set to zero
II) Scoring: negative scores are set to zero -> $S(i,j) = \max \begin{cases} 0 \\ S(i-1,j-1) + r(i,j) \\ S(i-1,j) + g \\ S(i,j-1) + g \end{cases}$
III) Traceback: from max entry to first zero

-> The alignment score is the sum of substitution scores and gap penalities. The alignment score reflects the aligned quality

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Substitution Matrix

DNA

	C	T	A	G
C	2	1	-1	-1
T	1	2	-1	-1
A	-1	-1	2	-1
G	-1	-1	-1	2

→ N x N goodness score matrix

→ Protein substitution Matrices:

→ 20 x 20

→ evolutionary substitution matrices

↳ frequency of amino acids

eg: PAM based on global alignments

BLOSSUM based on local alignments

Gap Penalties

→ Constant gap penalty: Each gap identical cost, independent of its length.

→ Goal: Min #Gaps

→ Linear gap cost: g·|length|, linear to size

→ Goal: Min Σ of # gaps & lengths

→ Affine gap cost: unlikely many small gaps

→ one big gap

↳ opening penality, $\sigma < 0$, extension penalty, $\epsilon < 0$

↳ $g = \sigma + (\lambda - 1)\epsilon$

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BLAST = Basic Local Alignment Search Tool

→ finds regions of local similarity between bio seq's

→ Alignments are assigned a significance score, i.e. E-Value

→ E-Value ☹

The number of different alignments with score equal or higher than a $core S that one can expect to occur in a DB search by chance.

↳ The lower E-value the more significant the score S

	Query		Subject
blastn	N—	→	—N
blastp	P—	→	—P
blastx	N≡	→	—P
tblastn	P—	→	≡N
tblastx	N≡	→	≡N

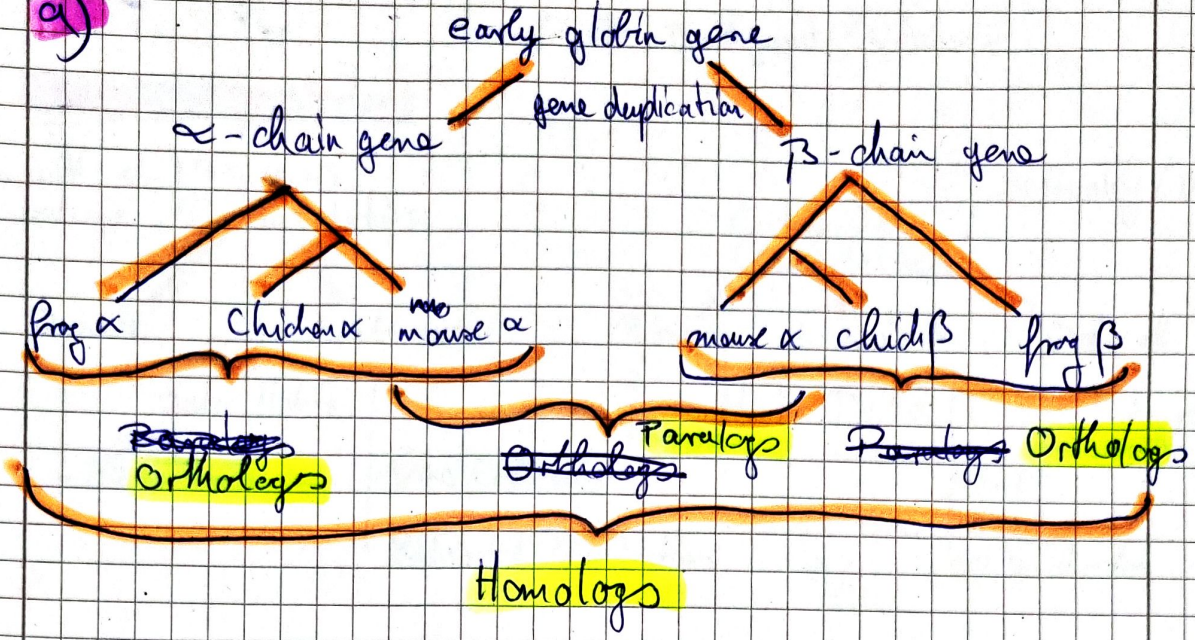
E value = 0 → no gaps

E value > 0 → gaps

GSB → Ex 5

(3)

a)



b)

Needleman Wunsch Alg used for global alignment.
Smith-Waterman Alg used for local alignment.
Both are examples for dynamic programming

GrSB

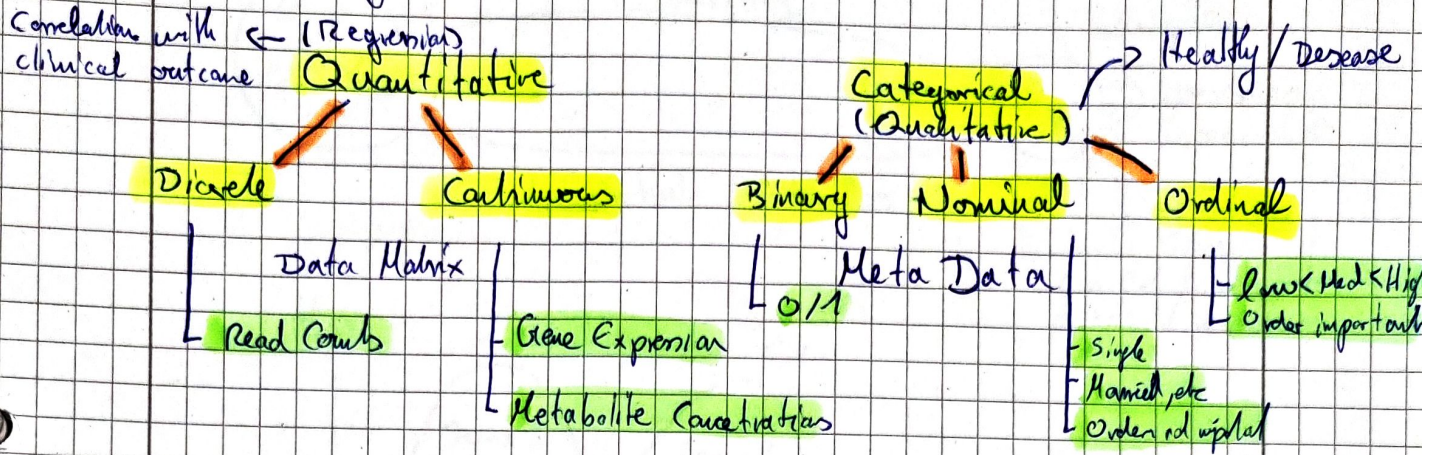
Biostatistics

q Madi

(C=> Forgroud / Signal

CC=> Background / Noise

Describing Data



-> Data are the observed values of a variable

-> variable is some characteristic of a population (e.g. Gene)

-> Values of the variable are the range of possible values for a var

-> Dimension of data are based on the # of variables it contains

-> Univariate : Measuring one variable per subject

-> Bivariate : Measuring two variables per subject

-> Multivariate : Measuring many variables per subject

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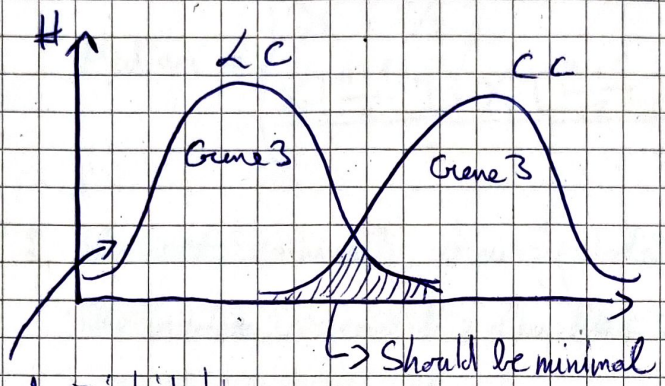
Key concept in statistics

↳ Diff sample can lead to diff results.



↳ Should be minimal

Normal Distribution

How do we describe the data

-> Central Tendency - center of the data location
  ↳ Mean, Median, Mode

-> Variability - the spread of the data
  ↳ Variance, Standard deviation

-> Relative Standing - distribution of data within the spread
  ↳ Quantiles, Range, IQR

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Mean, Median and Mode

→ try to summarize the whole data set in a single number

- **Mean:** the sum, divided by the number of data points

$$\text{Mean} = \frac{\sum X}{N}$$

- **Median:** the middle number of all ordered data points

$$\text{Median} = \begin{cases} X\left[\frac{n}{2}\right] & , n \text{ is even} \\ \frac{X\left[\frac{n-1}{2}\right] + X\left[\frac{n+1}{2}\right]}{2} & , n \text{ is odd} \end{cases}$$

- **Mode:** the most commonly occuring data point in a dataset. The modes is useful when there are a lot of repeated values in a dataset.

Variance, SD & SEM

- **Variance:** Ø of the squared distance to the mean
  ↳ not meaningful

$$\sigma^2 = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})^2$$

- **SD (Standard Deviation):**

$$\sigma = \sqrt{\frac{\sum_{i=1}^{n} (x_i - \mu)^2}{N}}$$

- **SEM (Standard Error of the Mean):**
  ↳ precision of the mean

$$SD_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

- **Covariance:** $\text{cov}(X, Y) = \frac{1}{n} \sum_{i=1}^{n} (x_i - E(x))(y_i - E(y))$
  ↳ A measure of how two vars change together

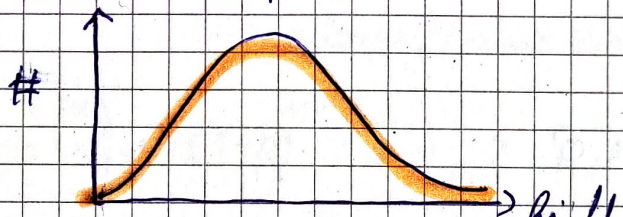
GSS

Quartile vs Quantile

- x% - Quantile: that value k where x% of the values are smaller than k and 100-x% of the values are bigger than k.
- Quartiles: are a special case of quantiles: 25% - / 50% - / 75% - Quantile

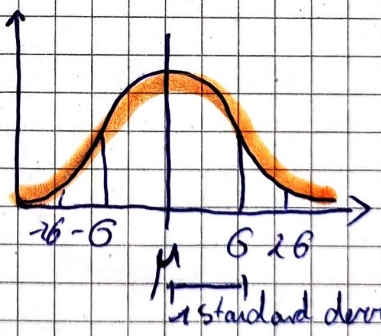
Univariate Statistics

-> compare different means, assume equal variance
-> single variable
-> plot frequency of values of that variable:



↳ Bell Curve, Gaussian / Normal Distribution

Standard Deviation



Z-Score:

How many SD away from the mean?

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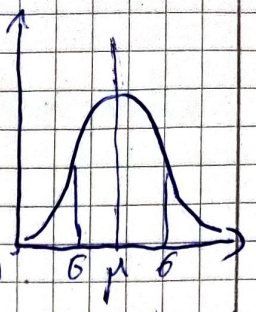
Normal Distribution



-> symmetric

-> Mean value (μ) at the center

-> Characteristic width called standard deviation (σ)

-> Most sets of biological or physical measurements -> normal distribution

-> Large set of small independent measurements display an approximate normal distribution -> Central Limit Theorem

Binomial Distribution

eg: How many coin flips show heads?

-> fixed number of repeated, identical, independent trials.

n is usually # of trials.

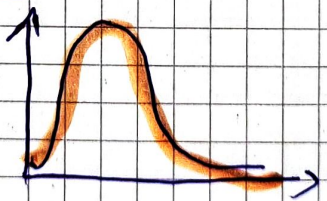
↳ success / failure with p for success

$$\Rightarrow P_{n,p}(X) = \binom{n}{x} p^x (1-p)^{n-x} \quad ; X = [0, n]$$

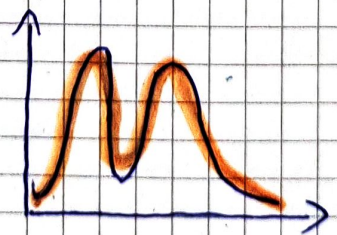
$$E(X) = np \quad ; \quad Var(X) = np(1-p)$$

Different Distributions

Unimodal



Bimodal



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Q-Q plot

-> Determine if two datasets come from populations with a common distribution.

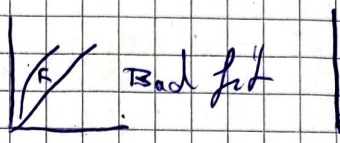
-> One against the other

-> 45 degree line:

If two sets come from a population with the same distribution, the points should fall approximately along this line

else

-> from different population



-> Similiar to probability plot

↳ the quantiles for one of the data samples are replaced with the quantiles of a theoretical dist.

Advantage:

(+) Sample sizes do not need to be equal

(+) Many distributional aspects can simultaneously tested.

eg: shifts in location, shifts in scale

GSB

Statistical Testing

- Hypothesis = statement that you can test
- Test = experiment → True / False
- Statistical Hypothesis: A probability of whether a statement is true / false can be given.

↳ Alternative Hypothesis (H_1 or H_A)

→ The opposite: Null Hypothesis (H_0)

→ H_1 & H_0 are mutual exclusive

eg:

H_1: Mice treated with drug X live longer:

$$E(T_X) > E(T_{untreated})$$

H_0: - II - do not live longer:

$$E(T_X) \leq E(T_{untreated})$$

=> We want to test if can reject H_0 and accept H_1

Principle of statistical testing

I) Find "test statistic", function modeling the distribution

II) Compare the observed values in the sample with this dist.

- Test Decision:

  large deviation: H_0 reject, H_1 accept

  small deviation: H_0 can not be rejected

  H_1 can not be accepted

BG: survival time without treatment

FG: survival time with treatment

BRUNNEN

GrSB

Error in tests

Wrong Decision:

-> Sample size too small

-> Sample, incidentally, could not be representative of the population

-> Test assumption not met

Type I error:

H_1 falsely accepted.

False positive (FP)

Type II error:

H_1 falsely rejected.

False negative

=> Reduction of Type I error increase Type II error and vice versa

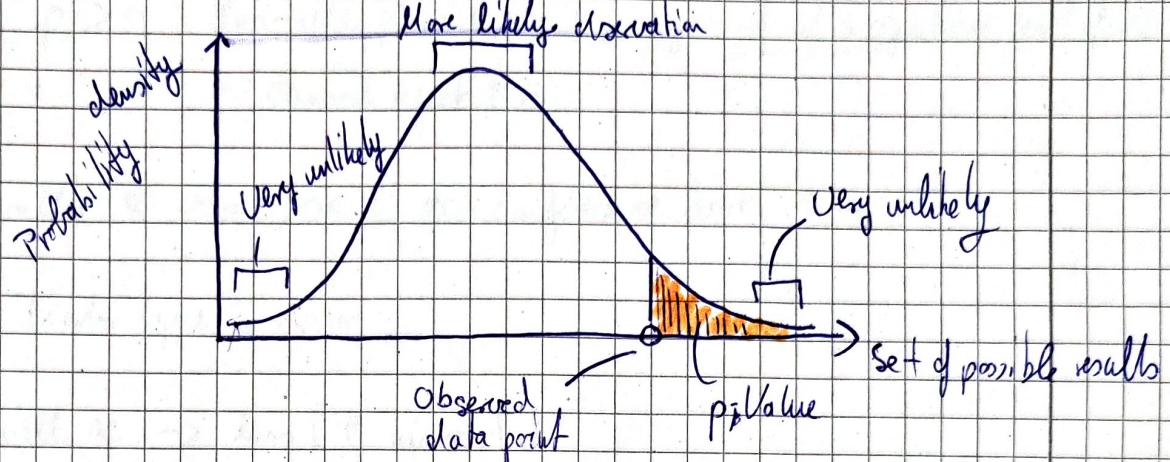
- **Sensitivity:** $Sens = \frac{TP}{TP + FN}$
- **Specificity:** $Spec = \frac{TN}{TN + FP}$
- **Accuracy:** $acc = \frac{TP + FN}{n}$

GASB

P-value

P-value : Probability of type I error (FP)

A p-Value is the probability of an observed result assuming that the null hypothesis is true.

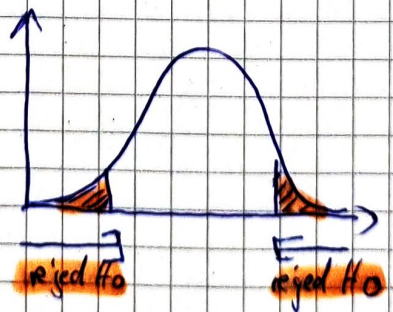


One-sided vs Two-sided hypothesis

- One sided hypothesis , deviation in one direction are considered
- Two sided hypothesis , deviation in both direction are considered

One Sided

Two Sided



GSB Statistic

Statistical Testing

significance level α (cut-off P-value)

- If $p \leq \alpha$: deviation from H_0 is significant at a significant level of α
  ↳ reject H_0
- If $p > \alpha$: deviation from H_0 is not significant at a significant level of α
  ↳ cannot reject H_0

usually $\alpha = 0,05$ or confidence level $1 - \alpha$

- α limits type I error

Small α -> less FP more FN

Large α -> more FP less FN

G SD — 6x G

1) Errors in Test

a)

		Reality	
		True	False
Measured	True	TP	FP (type I)
	False	FN (type II)	TN

b)

		Gold Stadd	
		True	False
Screening Tests	True	35	30
	False	15	120

Sensitivity: $sens = \frac{TP}{TP+FN} = 0,7$

Specificity: $spec = \frac{TN}{TN+FP} = 0,8$

Accuracy: $acc = \frac{TP+FN}{n} = \frac{35+120}{200} = 0,775$

2) z-score

$\mu = 60$, $\sigma = 8$, meanA = 62.7 for $n = 25$ people

a) $z = \frac{62.7 - 60}{8} = 0,3375 = \frac{\bar{x} - \mu}{\sigma}$

b) H_1: People who smoke will have higher score "health problem"

H_0: People who smoke will have not high score in health problem

→ 62,7 is 0,3375 Standard Deviations above the mean.

is close to the mean score.

→ accept H_0 and reject H_1